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Optimal Transport-Based Gradient Operators for Seismic Imaging: A Robust Approach for Subtle Structural Detection

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Abstract

In the complex landscape of seismic data, fault detection is the key to unlocking the secrets of geological structures and pinpointing oil and gas reservoirs, but traditional gradient operators like finite difference and Sobel filters often fall short, as they are highly susceptible to noise and struggle to reveal subtle structural features. Our innovative method, grounded in optimal transport theory, revolutionizes fault detection by leveraging Wasserstein distances to compute gradients in seismic imaging, envisioning seismic images as 2D mass distributions and calculating gradients based on the minimal transport cost required to align neighboring patches, making it inherently robust against noise and expertly identifying small structural changes, such as faults, fractures, and stratigraphic boundaries. With the incorporation of entropic regularization ensuring numerical stability and GPU acceleration empowering efficient processing of large datasets, our method outperforms traditional approaches in both synthetic and real datasets, demonstrating its superiority in fault detection and offering a reliable, practical solution for large-scale seismic imaging applications, ultimately unveiling hidden faults and providing a clearer understanding of the geological landscape.

Introduction

Seismic imaging and seismic attribute computations are crucial for characterizing reservoirs, especially for identifying small faults and stratigraphic boundaries. However, traditional amplitude-based gradient operators often fall short in detecting these features due to noise, unclear amplitude changes, and complex geology. These limitations hinder the accuracy and reliability of seismic interpretations, which are crucial for oil and gas exploration.

Edge detection is a fundamental aspect of image processing, aimed at simplifying image data to minimize the amount of information that needs to be processed (Canny, 1986). Ideally, edge detection should result in a set of connected curves that indicate the boundaries of objects, surface markings, and changes in surface orientation. This process not only highlights the structural properties of an image but also filters out less relevant details, making it easier to focus on significant features.

In the context of seismic attributes, the introduction of optimal transport theory, particularly Wasserstein distances, offers a promising solution to the challenges posed by traditional methods. By treating seismic

images as 2D mass distributions, this approach calculates gradients based on the minimal transport cost required to align neighboring patches. This method demonstrates improved resistance to noise and enhanced sensitivity to small structural changes, making it more effective in detecting subtle faults and stratigraphic boundaries.

The integration of entropic regularization ensures numerical stability, while GPU acceleration enables efficient processing of large datasets, which is essential for practical applications in the oil and gas industry. The results from both synthetic and real datasets underscore the method's capabilities and reliability, suggesting that this approach could significantly advance fault detection and reservoir characterization. This innovation has the potential to provide more accurate and detailed insights into subsurface structures, thereby supporting more informed decision-making in exploration and production efforts.

Theory and Background

Understanding the Limitations of Traditional Gradient Operators

Traditional gradient operators are essential tools for detecting changes in amplitude, which is crucial for identifying features like faults in seismic data (Al-Dossary, 2015). However, these operators have some limitations. The main issues are:

1. **Sensitivity to Noise:** Gradient operators can be easily affected by random fluctuations in the data, leading to false detections. This is especially problematic in high-noise environments where important geological features can be obscured.
2. **Complex Geological Settings:** Geological features are often not simple or linear; they can be curved, fractured, or vary in scale. Traditional gradient operators might assume a level of simplicity or linearity, leading to misinterpretations (Al-Dossary and Al-Garni, 2013).
3. **Mathematical Constraints:** These operators rely on linear operations, which may not handle non-linear relationships in geological data well. Also, real-world noise is often non-Gaussian, which linear operators are not good at managing (Aqrawi and Boe, 2011).

To overcome these challenges, more advanced techniques are necessary

1. **Lineament Preserving filters:** Al-Dossary and Marfurt (2007) introduced novel filtering techniques designed to preserve lineaments—linear or curvilinear features indicative of geological structures like faults and fractures—in seismic data. Traditional filtering methods often distort or obscure these features, which are vital for accurate geological interpretations. The techniques are applied prior to computation of seismic attributes to reduce the noise while preserving the structures. Al-Dossary (2015) discussed methods for enhancing the quality of seismic data to improve the detection of channels by amplitude adjustments, to make channel detection more accurate and reliable.
2. **Adaptive Multi-scale and Multi-direction Approaches:** Techniques like wavelet transforms and directional amplitude gradient and adaptive Soabelt can capture features at various scales and directions, reducing the effect of small-scale noise (Aqrawi, 2014) and (Al-Dossary and Al-Garni, 2013).
3. **3D Operators:** Applying gradient operators in three dimensions can preserve the continuity of geological features (Ashraf et al., 2016).

By adopting these advanced methods, the limitations of traditional gradient operators can be effectively addressed, leading to more accurate and reliable seismic interpretations.

Optimal Transport Theory: A Novel Approach for Geophysical Data Analysis

In the realm of geophysical data analysis, traditional gradient operators have been the cornerstone for identifying edges and changes in amplitude. However, these operators face significant challenges when dealing with noisy data and complex geological settings. Their sensitivity to noise, assumption of non-linear geological features, and the difficulty of extending some of them to three-dimensional data pose considerable challenges. To overcome these limitations, Optimal Transport Theory emerges as a promising alternative.

Understanding Optimal Transport Theory

Optimal Transport Theory is a mathematical framework that measures the minimal cost required to transform one distribution into another. Essentially, it seeks the most efficient way to rearrange mass from one distribution to match another. This concept is particularly useful in comparing and aligning different datasets, making it a valuable tool in data analysis.

Application in Geophysical Data Analysis

In the context of geophysical data analysis, Optimal Transport Theory can be applied to address the shortcomings of traditional gradient operators in several ways:

1. **Noise Robustness:** By focusing on the overall distribution of data rather than local changes, Optimal Transport Theory can mitigate the impact of noise. It provides a more global perspective, reducing the likelihood of false detections caused by random fluctuations.
2. **Handling Complex Geological Features:** Geological features are often non-linear and multi-scale, such as fractures, folds, and channel systems. Optimal Transport Theory can handle these complexities by considering the global structure of the data, allowing for a more accurate identification of intricate geological features.
3. **Three-Dimensional Data Management:** Seismic data is inherently three-dimensional, and Optimal Transport Theory can be extended to handle 3D distributions. This capability ensures that the continuity of geological features is preserved, offering a more comprehensive understanding of subsurface structures.
4. **Distribution Alignment:** In scenarios where data from different sources or scales is available, Optimal Transport Theory can align these distributions, facilitating a more integrated analysis. This is particularly useful in multi-scale studies where features vary significantly in size and orientation.
5. **Denoising and Feature Enhancement:** By transforming data into a space where geological features are more distinguishable, Optimal Transport Theory can enhance the clarity of these features. It can also help in denoising by aligning the data with a reference distribution that represents the expected geological signal.

Optimal Transport Theory (Peyré and Cuturi, 2019) offers a novel and robust approach to geophysical data analysis, addressing the limitations of traditional gradient operators. Its ability to handle noise, complex geological features, and three-dimensional data makes it a valuable tool for enhancing the accuracy and reliability of geological interpretations. As the field continues to evolve, the integration of Optimal Transport Theory with other advanced techniques promises to unlock new insights into the Earth's subsurface, aiding in more informed decision-making in exploration and production.

The Wasserstein distance measures the minimal cost to change one distribution into another. In seismic imaging, images are treated as 2D mass distributions, with pixel amplitudes showing mass density. The gradient operator is defined as the rate of change of the Wasserstein distance between neighboring patches. This approach is better at handling noise and finding small changes compared to traditional amplitude-based methods.

The Sobel filter (Aqrawi and Boe, 2011), (Al-Dossary and Al-Garni, 2013), (Aqrawi, 2014) and (Ashraf et al., 2016) a popular edge-detection tool, uses 3×3 kernels to find horizontal and vertical edges. While effective, it can be affected by noise and may miss small changes. The Wasserstein-based operator has advantages in being robust, sensitive, and mathematically sound but needs more computing power. GPU acceleration and entropic regularization help with these challenges, making it possible to process large datasets efficiently.

Methods

1. **Seismic Image Representation:** Treat the image as a 2D mass distribution using a softmax transformation to normalize amplitudes into a probability density function (PDF).
2. **Wasserstein Distance Calculation:** Calculate the Wasserstein distance between local patches (e.g., 5×5 or 7×7) and their neighbors. The distance shows the minimal transport cost.
3. **Gradient Calculation:** Calculate horizontal (G_x) and vertical (G_y) gradients as differences in Wasserstein distances between patches. The total gradient is the absolute sum of G_x and G_y .
4. **Entropic Regularization:** Add an entropy penalty term to ensure numerical stability and smoothness in calculations.
5. **GPU Acceleration:** Use parallel processing on GPUs to efficiently handle large datasets.

Results

- **Synthetic Dataset:** The Wasserstein-based operator successfully captures faults and stratigraphic boundaries in noisy conditions, performing better than traditional methods.
- **Real Dataset:** The method shows better continuity and fewer false positives in detecting small faults and boundaries in complex reservoir settings.

Examples of Channel and Fault Enhancements

In this section we will show the new algorithm usefulness to reveal channels and faults in seismic data. Edge detection algorithms are commonly used in the processing of 2D and 3D seismic data to extract important but subtle geological features such as faults and channels, which often appear as sharp edges and are not easily detected in conventional seismic amplitude data.

Figure 1a shows our first test case, an amplitude time slice extracted from a 3D seismic cube. Noticeable in the figure are two diagonally running channels. Main channel indicated by red arrows and secondary channel indicated by blue arrows.

We now we show the result of applying edge detection to the input respectively. Figures 1b (edge detection using Sobel Operator) and figure 1c (Edge detection using the new method) we show the result of applying edge detection to the input respectively. The main channel is more visible in Figure 1c and secondary channels become more visible.

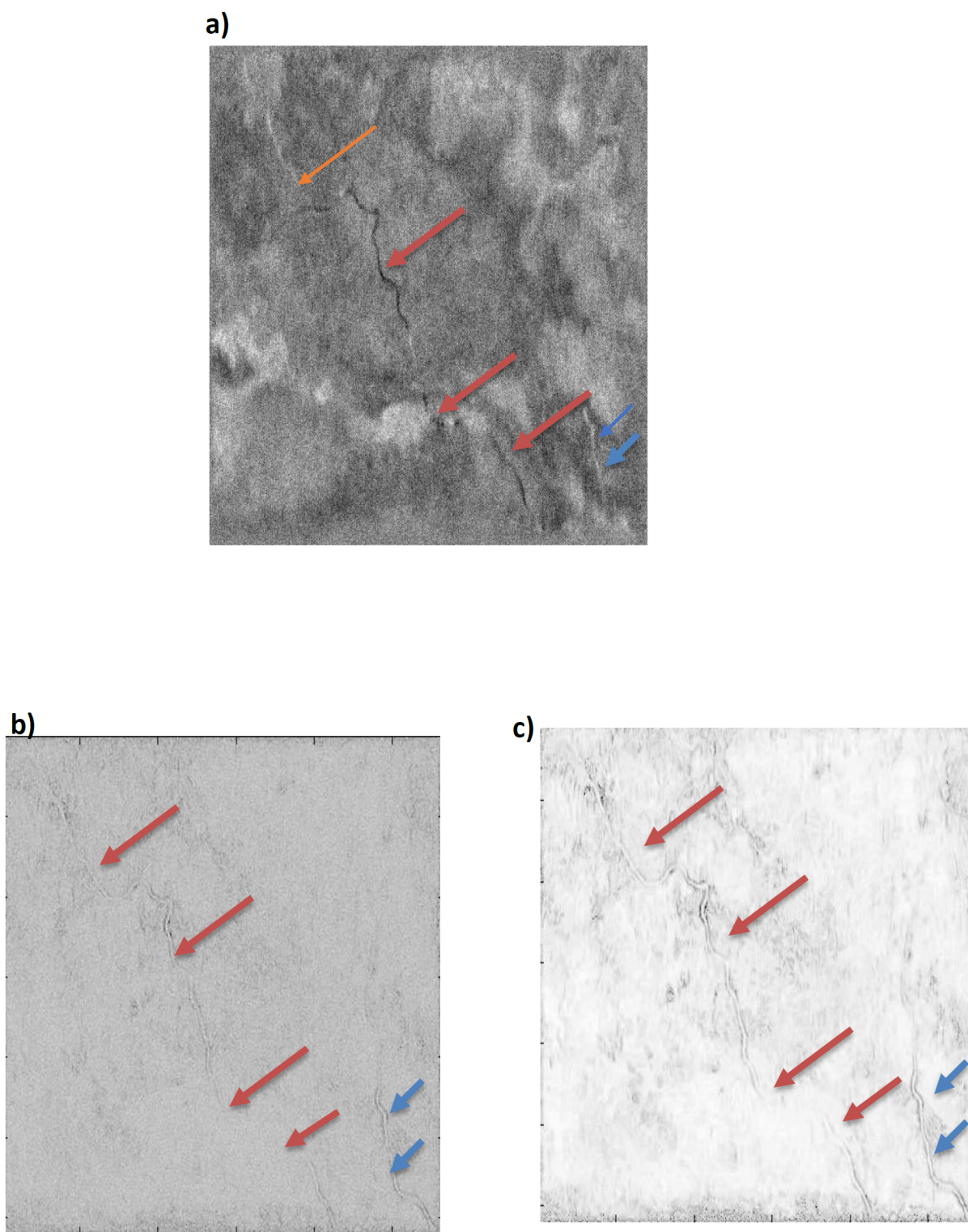


Figure 1—Input time slice from a 3D seismic volume (a) An amplitude time slice b) corresponding time slice using the Sobel edge-detection algorithm. (c) The corresponding time slice through edge-detection using the new algorithm

Next, we show the results of applying the proposed method to noisy volume from a different 3D seismic data. [Figure 2a](#) shows the input vertical amplitude section extracted from the 3D seismic volume. In [figure 2b](#), we display the edge detection result using the Sobel operator run on the input data. The faults are hard to identify. In [figure 2c](#) we show the result of the proposed edge detector applied to the seismic data after. Data is cleaner and the faults are preserved.

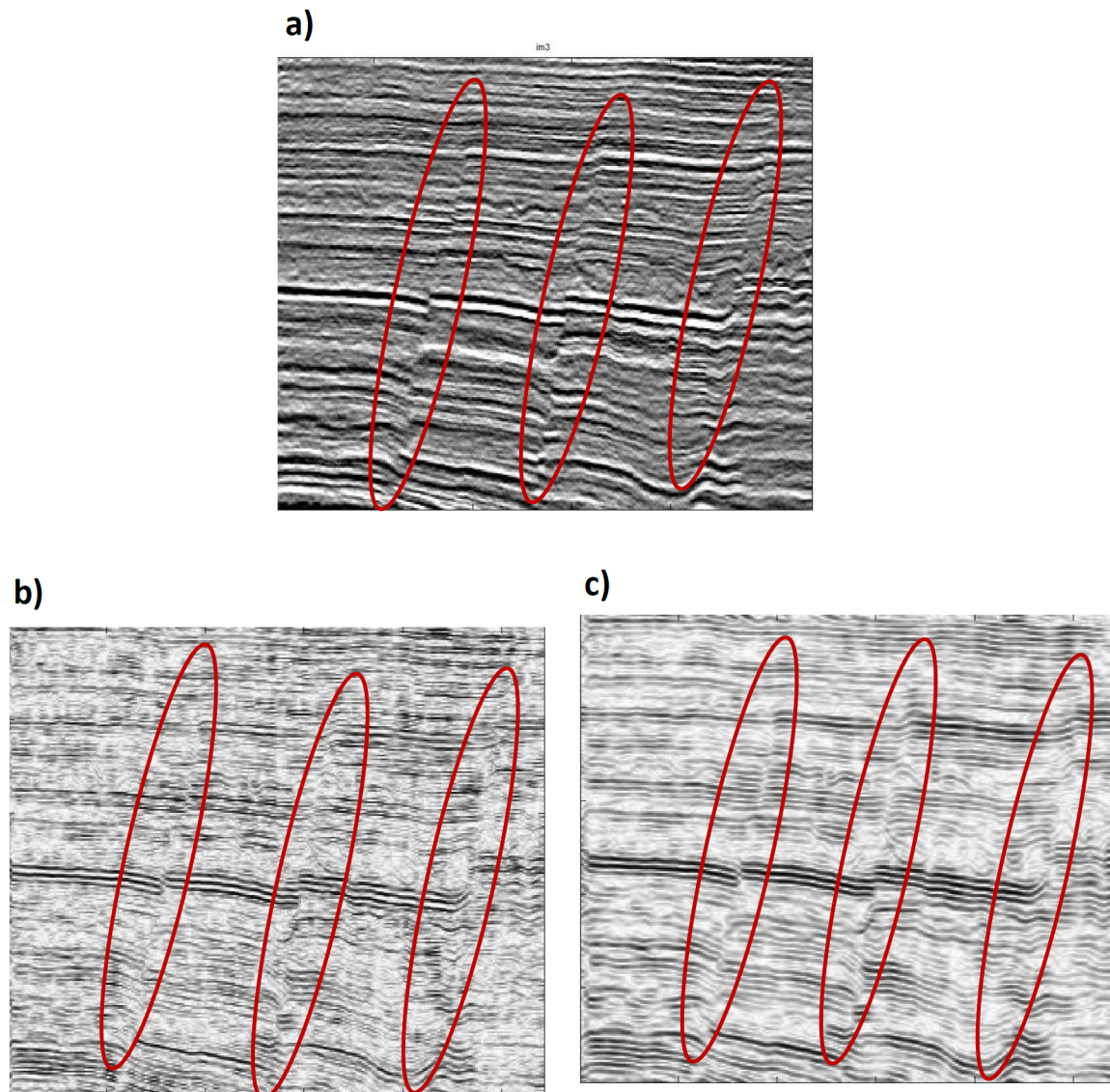


Figure 2—(a) Vertical slices through the seismic volume. (b) - (c). The corresponding Vertical slices after edge detection. (c) using Sobel edge detection and (c) after the proposed method. The proposed method shows a reduction in high frequency noise and improved resolution of all the faults.

Edge Detection in Seismic Data: Insights from a New Algorithm Based on Wasserstein Distances

The new edge detection algorithm, which utilizes Wasserstein distances, has demonstrated its effectiveness in unveiling geological features such as channels and faults within seismic data. This is crucial as these features are often subtle and not easily discernible in traditional seismic amplitude data. The algorithm's performance was tested on two datasets: a clean amplitude time slice and a noisy vertical amplitude section. In the clean dataset, the new method was compared to the widely used Sobel Operator. The results showed that the new algorithm not only highlighted the main channels more distinctly but also brought secondary channels into clearer view. This superior performance in detecting both primary and secondary features underscores its potential in enhancing geological interpretations.

In the noisy dataset, the algorithm's ability to clean the data while preserving faults stands out as a significant advancement. Traditional methods often struggle with noise, which can obscure important geological details. However, the new method's robustness against noise ensures that faults remain identifiable, maintaining the integrity of the data crucial for geological analysis. This breakthrough is

particularly beneficial for hydrocarbon exploration and subsurface studies, where accurate detection of geological features can lead to more informed decision-making. While the algorithm shows great promise, considerations such as computational requirements and ease of implementation will be important factors in its practical application. Overall, this innovation represents a notable improvement in seismic data analysis, offering clearer insights into geological structures and enhancing our understanding of the subsurface.

Discussion

The Wasserstein-based operator is strong in handling noise, finding small changes, and being mathematically sound. However, it faces challenges in computational complexity, parameter sensitivity, and extending to 3D datasets. Future work could look into alternative regularization techniques, automatic parameter tuning, and 3D applications.

The introduction of optimal transport-based gradient operators in seismic imaging marks a significant advancement in the field, with implications that extend far beyond the case studies presented. This method not only addresses the limitations of traditional amplitude-based gradient operators but also opens new avenues for image processing in geophysics and beyond.

Theoretical Contributions

The method represents a substantial theoretical advancement in seismic imaging by integrating concepts from optimal transport theory, particularly Wasserstein distances, into gradient computation. By treating seismic images as 2D mass distributions, the approach provides a mathematically rigorous framework for capturing subtle structural features that are often obscured by noise. This innovation underscores the potential for theoretical advancements in image processing to revolutionize geophysical applications, encouraging further exploration of novel mathematical frameworks in the field.

Interdisciplinary Applications

The applicability of this method is not confined to seismic imaging. The challenges of noise robustness and sensitivity to subtle variations are common in various fields, such as medical imaging and remote sensing. The success of Wasserstein-based gradient operators in seismic imaging suggests their potential utility in these domains, where accurate detection of faint features is critical. For instance, in medical imaging, this method could enhance the detection of subtle anatomical structures or pathological features, improving diagnostic accuracy.

Industry Impact

In the petroleum industry, the ability to more accurately characterize reservoirs can lead to improved resource management and exploration efficiency. By providing more reliable data on subtle faults and stratigraphic boundaries, this method can reduce exploration risks and costs. Additionally, it can contribute to better decision-making in resource extraction, potentially leading to more sustainable practices by minimizing the environmental impact of exploration activities.

Technological Advancements

The method's leverage of GPU acceleration and entropic regularization highlights the importance of computational innovation in advancing geophysical techniques. This approach not only enhances processing efficiency but also demonstrates the synergy between advanced computational methods and geophysical applications. The integration of machine learning, particularly in automating parameter tuning and extending the method to 3D datasets, represents a promising direction for future research, underscoring the evolving role of artificial intelligence in seismic imaging.

3D Seismic Imaging

Extending this method to 3D datasets is a logical next step, with the potential to provide even greater insights into subsurface structures. The added complexity of 3D data could be managed through advanced computational techniques, building on the existing use of GPU acceleration. This extension could offer a more comprehensive understanding of geological features, enhancing the ability to identify complex fault networks and stratigraphic boundaries in three dimensions.

Integration with Other Technologies

The integration of this method with other advanced imaging technologies, such as machine learning-based approaches, represents a promising avenue for future research. Combining the robustness of Wasserstein-based gradient operators with the adaptability of machine learning could lead to even more sophisticated tools for seismic imaging, capable of handling the complexities of real-world datasets with greater efficacy.

Multi-Scale Analysis

Adapting the method for multi-scale analysis could provide insights at various resolutions, from detailed local features to broader regional structures. This adaptability would make the method versatile, suitable for a range of applications from detailed reservoir characterization to regional geological studies.

Conclusions

This paper introduces an innovative approach to gradient calculation in seismic imaging by leveraging Wasserstein distances, a concept rooted in optimal transport theory. By treating seismic images as mass distributions, the method computes gradients based on transport costs, thereby enhancing performance in noisy conditions—a common challenge in seismic data processing. The integration of entropic regularization ensures numerical stability, while GPU acceleration optimizes computational efficiency, making the approach suitable for large-scale applications. Comprehensive validation using both synthetic and real datasets demonstrates the method's robustness and effectiveness, providing a reliable solution for advancing seismic imaging techniques. This research not only bridges the gap between optimal transport theory and seismic interpretation but also enriches the reservoir characterization toolbox with a powerful new method, promising significant contributions to the field of geophysical data analysis.

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